include\wire_bus.h

```c
// Schnittstelle der Bus-/Leitungsfunktionen
#ifndef WIRE_BUS_H
#define WIRE_BUS_H

#include <Arduino.h>
#include <avr/io.h>
#include <stdint.h>
#include <stdbool.h>

// ------------------------------------------------------------
// wire_t – single-wire open-drain line descriptor
// each wire_t stores 3 infos
// port    : pointer to the AVR PORT struct (e.g. &PORTA, &PORTB)
//  pin_bm  : bitmask of the pin  (e.g. PIN1_bm)
//  pin_idx : numeric index 0-7, used to select PINnCTRL for pull-up
// ------------------------------------------------------------

typedef struct {
    PORT_t  *port;
    uint8_t  pin_bm;
    uint8_t  pin_idx;
} wire_t;

// ------------------------------------------------------------
// Wire primitives
// examples:
// line_release(const wire_t *w)  = function definition
// line_release(&upstream)       = function call with &upstream pin defined in wire_t as argument
// ------------------------------------------------------------

void wire_release(const wire_t *w); // releases the line (high level)
void wire_low(const wire_t *w);     // pulls the line low (low level)
bool is_line_high(const wire_t *w); // checks if the line is high (returns true if high)
```

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34
35 // ------------------------------------------------------------
36 // UART-like byte transfer
37 // ------------------------------------------------------------
38
39 void tx_byte(const wire_t *w, uint8_t b);  //transmission of byte
40
41 bool rx_byte(                              // receiving of byte
42     const wire_t *w,
43     uint8_t *out,
44     uint16_t timeout_us
45 );
46
47 bool rx_byte_data_frame(                   //receiving of message (sent bits) without start bit
48     const wire_t *w,
49     uint8_t *out
50 );
51
52 //------------------------------------------------------------
53 // Frame transfer
54 // Frame format:
55 // SYNC LEN PAYLOAD CHECKSUM
56 //------------------------------------------------------------
57
58
59 uint8_t calc_checksum(         //crc check
60     uint8_t payload_len,
61     const uint8_t *payload
62 );
63
64 void tx_frame(                  // transmitting data frame
65     const wire_t *w,
66     const uint8_t *payload,
67     uint8_t payload_len
68 );
69
```

```
70 bool rx_frame(                // receiving data frame
71     const wire_t *w,
72     uint8_t *payload,
73     uint8_t max_payload_len,
74     uint8_t *payload_len,
75     uint16_t timeout_us
76 );
77
78 #endif
```